

JAGRITI TRIPATHI

Software Engineer

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PROFESSIONAL SUMMARY

Software Development Engineer with 2 years of experience building scalable backend systems and enterprise financial applications at Nomura. Proficient in Java, Spring Boot, React, SQL, distributed systems, REST APIs, and DevOps practices. Experienced in performance optimization, observability, production support, and CI/CD, with a strong interest in AI, distributed systems, and modern software architecture.

WORK EXPERIENCE

Software Engineer, NOMURA (Jul 2024 – Jan 2026 | 1.5 years)

Project: Trade Distribution Platform (Middle Office)

- Built high-performance search dashboards, optimizing existing backend APIs in **Java/Spring Boot**, reduced response latency from ~2 minutes to ~20 seconds (by ~83%), achieved through query profiling, index restructuring.
- Developed broker onboarding workflows for a microservice that processes incoming broker files consisting up to 100k+ trades into standardized **PDM/XML** formats and automated file-summary deliveries using **Bash & Autosys**.
- Improved data consistency and validation across front-office trade processing pipelines, reducing downstream recon issues.
- Replaced SSH-based log triage with a centralised Grafana + Prometheus + OpenTelemetry observability stack, reducing incident investigation from manual host lookups to instant PromQL queries across logs, metrics, and service health.
- Provided **L3 production support**, performing end-to-end triage across inbound files, relational DBs, and TIBCO EMS queues, including DLQ analysis and message replay.
- Spearheaded complete rewrite of a legacy web application from React15 → React 18, reducing load time by ~80% (2.5 min → <30s) while enhancing UX, responsiveness, and cross-team usability; successfully adopted in production.
- Configured and integrated **NGINX** as a reverse proxy for web service, enabling frontend and backend services to be served under organization-specific context paths for intranet deployment while tuning gateway timeout policies to prevent long-running DB requests.
- Migrated containerized microservices from **Docker Swarm (RHEL7)** to **Podman (RHEL8.10)** using **Quadlets and rootless systemd units** for stable enterprise deployments.
- Authored and maintained **JUnit/Mockito tests**, improving overall code reliability by achieving ~85% code coverage on SonarQube, strengthening **GitLab CI/CD** pipelines.

Technology Intern (Developer), NOMURA (Jan 2024 – Jun 2024 | 6 months)

- Designed and developed a modern web frontend end-to-end using **React 18**, upgrading legacy UI components.
- Integrated third-party APIs using **Spring Boot** for secure data retrieval for the Trade Service platform.
- Worked on a mini **web-scraping** project to extract and process structured data.

EDUCATION

B.Tech in Electronics and Communication Engineering - CGPA: 7.47

Indira Gandhi Delhi Technical University for Women, Delhi

2020-2024

XII (PCM, CBSE) - Percentage: 90.2%

School Of Excellence, Rohini

2018-2020

TECHNICAL SKILLS

Languages & database: Java, Python, JavaScript, TypeScript, Pydantic, Bash, XML/XSLT, PostgreSQL

Backend: Spring Boot, Apache Camel, Microservices, REST APIs, Event-Driven Architecture, Tibco EMS

Frontend: ReactJS, Material-UI, Bootstrap, TailwindCSS

GenAI: Prompt Engineering, LLMs, RAG, Vector Embeddings, VectorDBs(Chroma/FAISS), Hugging Face, PyTorch

DevOps & Infra: Docker, Podman, GitLab CI/CD, Linux, Autosys, Postman, NGINX

Observability & Testing: Grafana, Prometheus, OpenTelemetry, JUnit, Mockito

Concepts: Distributed Systems, Low-Latency Design, SOLID, Design Patterns, DSA, Agile

PROJECTS

UnderwriteLM [Python | Pydantic | LLM APIs | Prompt Engineering | FastAPI | Docker]

Developed an LLM-powered underwriting decision system with prompt-engineered guardrails, synthetic case generation, and an LLM-as-judge evaluation framework; served via FastAPI and Docker.

GetMyTix [Spring Boot | ReactJS | PostgreSQL]

Built a high-throughput ticket booking system designed to handle intense concurrent demand, leveraging optimistic locking, distributed conflict control, and real-time seat state orchestration to prevent overselling and maintain booking integrity.

SourceSea [React | Typescript | NodeJS | Express | SQLite]

Developed an end-to-end resource management platform implementing layered architecture, reusable components, and centralized error handling to resolve bookmark loss for important webpages.

ACHIEVEMENTS

- Databricks Accredited Generative AI Fundamentals
- Pursuing IBMxGFG Gen-AI course.
- Certified for 30 Days of Google Cloud (Performed 70 assigned labs.)
- 2nd Runner-up in LeanIn Hacks (Created a disability toolkit plugin for CWSN)